



PROPOSAL

Lake Minnehaha Remediation Plan Phase 2

City of Cheyenne

RFP No. 7-15

January 2015

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AVRES
ASSOCIATES

January 16, 2015

Mike Vinson
City of Cheyenne, City Engineers Department
2101 O'Neil Ave., Room 206
Cheyenne, WY 82001

Re: Technical Proposal – Lake Minnehaha Remediation Plan Phase 2

Dear Mr. Vinson:

Ayres Associates is excited that the City of Cheyenne is moving ahead with Phase 2 improvements for the Lake Minnehaha Remediation Plan. Our staff members have enjoyed working with you on the plan and the Phase 1 improvements, and we are ready to go to work with you to complete Phase 2. Here's what we offer the City:

- **Knowledge and commitment.** Because we have worked with the City on so many municipal, water resources, and transportation projects over the years, we are tuned in to your preferences. We know your staff, goals, and procedures. We also will be able to incorporate our past successes from other similar projects to improve the Holliday Park area for the community to enjoy for many years to come.
- **Familiarity.** Our previous experience on past Lake Minnehaha and Holliday Park projects will allow our team to begin work immediately. We come with a solid communications plan, ready to keep all stakeholders well-informed on the design of your project.
- **Full-service team.** Not only does our team have the technical expertise to provide a design to meet necessary specifications, but our subconsultants – Steil Surveying Services and Terracon – are familiar to the City and Ayres Associates. This same team has worked on many projects together and will provide a seamless group of professionals to bring your project to a successful completion.

The signers of this proposal shown below declare that the companies interested in this contract as principals are named herein, specifically Ayres Associates, Steil Surveying Services, and Terracon. This proposal is offered in all respects as a fair proposal, in good faith without collusion or fraud. James Schall, PhD, PE, has authority on behalf of Ayres Associates to bind the company at the contract value shown in the price proposal.

Thank you for this opportunity to present our proposal. We look forward to working with you to improve Holliday Park and the Duff Avenue corridor. If you have any questions or need more information, please contact me.

Sincerely,

Ayres Associates Inc

Christopher G. Pletcher, PE
Project Manager
307.634.9888
PletcherC@AyresAssociates.com

CGP:ejw

James D. Schall, PhD, PE
Vice President
307.634.9888
SchallJ@AyresAssociates.com



Project Team

Ayres Associates will provide an experienced team to the City of Cheyenne for your Lake Minnehaha Remediation Plan Phase 2 project. Many of these team members will be familiar to you from past projects. Our team of professional engineers, technicians, and support staff has successfully completed many similar projects, including several for the City of Cheyenne. We know how to work with you to complete this project efficiently.

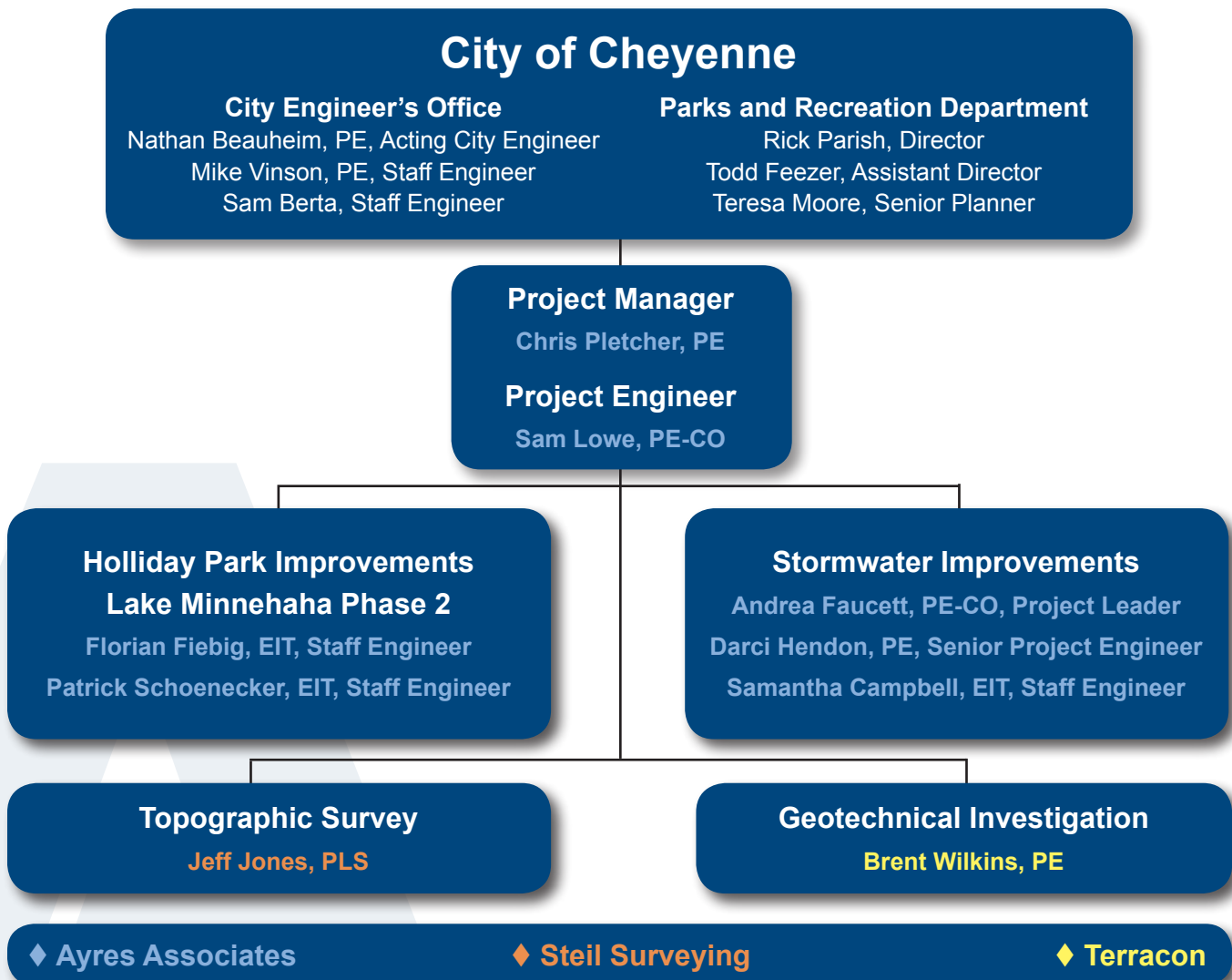
Chris Pletcher, PE, will serve as project manager. He will work out of our Cheyenne office as needed and will be your primary contact, attend meetings, schedule work, monitor progress and budgets, and see that this project is completed to your

satisfaction. Other Ayres team members are located in our Cheyenne or Fort Collins offices. Our project team's roles and structure are outlined below.

Subconsultants

Ayres is excited to work again with Steil Surveying Services and Terracon as subconsultants for this project. Ayres has worked with these firms on several recent projects, most recently the Board of Public Utilities' Crow Creek North Interceptor Sewer Recommissioning project, which is out for bid.

Resumes showing relevant experience for our project team members follow in this section.



Chris Pletcher, PE
Project Manager



Total Experience

18 Years

Registrations

*Registered Professional Engineer,
WY, 2007; CO, 2001; UT, 2014;
IA, 2010; NE, 2012*

Certifications

*Confined Space Training, Ayres
Associates, 2014*

Education

*BS, Civil Engineering, Clarkson
University, 1996*

*BS, Physics, Roberts Wesleyan
College, 1995*

Memberships

American Public Works Association

American Water Works Association

*Colorado Association of Stormwater
and Floodplain Managers*

Water Environment Federation

Chris joined Ayres Associates in 2006, bringing 10 years of experience in civil engineering analysis, design, and construction. He has comprehensive, substantial experience in corridor-type utilities, including storm drainage and sanitary sewer and water systems. He provides computer-based demand forecasting, master planning, capacity sizing, preparation of detailed construction plans, review of plans on behalf of local municipalities, and construction inspection services.

This is what Chris Matkins, water utilities manager for the City of Loveland, Colorado, said about Chris' work on wastewater flow monitoring projects: "Ayres Associates works closely with our operations crew; we have a high degree of trust in them. In fact, the trust is so high that Chris is almost considered an extension of our staff; he is able to make decisions and changes on the fly."

Related Projects

Lake Minnehaha Remediation Plan, City of Cheyenne: Responsible for project management. Project involved developing plan for ecological remediation of historic Lake Minnehaha in Holliday Park. Lake has high phosphorus and nitrogen levels that promote phytoplankton blooms, pH levels higher than most aquatic species can tolerate, water temperatures that exclude cold water fish, odors that indicate anoxic conditions in sediment, and limited depth. Tasks included water quality sampling, sediment sampling, locating site for disposal of dredged materials, environmental analysis, and preparation of remediation plan.

Remediation of Historic Lake Minnehaha Holliday Park – Phase 1, City of Cheyenne: Responsible for project management. Project involved taking concepts and action points from remediation plan to full design. Items suggested in remediation plan were divided into Phase 1 tasks, which were primarily within boundary of Lake Minnehaha, and Phase 2 tasks, which were outside of lake. Phase 1 implemented deepening of lake, removal of island, upgrading lake's outlet structure, and outlet pipe and riprapping bank line.

Capitol Basin Storm Drain Assessment, City of Cheyenne: Responsible for project management. Project involved confined space entry into an 84-inch storm drain in City's historic downtown area to try to determine source of odors. Project team also collected water quality samples from storm drains upstream.

Concrete Trail and Culvert Design, Town of Milliken, CO: Responsible for overall direction of project, CDOT approval, and coordination with client. Project involves designing concrete trail for Town of Milliken, including a ditch crossing. Services include developing plans and specifications to construct a concrete trail for project being funded through grant from the Colorado Department of Transportation.

Crow Creek North Interceptor Sewer Rehabilitation, Cheyenne Board of Public Utilities:

Responsible for project management. Project involves design and construction services for rehabilitation of 7,000 feet of 30-inch to 36-inch concrete pipe sanitary sewer that has been offline for several years. Project challenges include groundwater infiltration, new flow diversion structure, 300 feet of new pipe, and installation of liner into existing pipe. Design effort includes reviewing closed circuit TV footage, confined space entry visual inspection, survey, potholing, groundwater review and modeling, new manholes, and coordination with private property owners.

East Pershing Corridor Design and Construction, City of Cheyenne:

Project manager for sewer and water design and installation. Project involved providing roadway and utility design and construction engineering for reconstruction of East Pershing Boulevard from Dunn Avenue to Converse Avenue, including reconstruction of East Pershing/Converse Avenue/19th Street intersection as five-leg, multi-lane roundabout. Project included public outreach, right-of-way survey, plan for phased construction to minimize traffic disruption, design of modular-block and cast-in-place concrete retaining walls in front of homes, and design of cantilevered sign structures directing traffic to correct lanes at roundabout. Also included were reconstruction of portions of Pershing Boulevard, replacement of water mains and associated valves and hydrants, replacement of six sanitary sewer manholes and 490 linear feet of sanitary sewer, installation of two new manholes, and relining or replacing additional 485 feet of sanitary sewer.

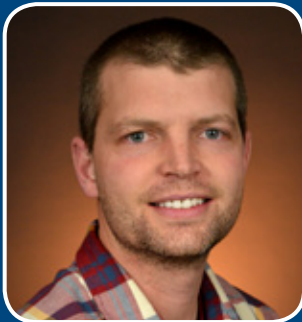
Lower Dry Creek Wetland, Laramie County

Planning & Development Office: Responsible for water quality sampling. Project involved engineering services for 2.5-acre forebay pond with 2.5-acre

wetland chase to provide sediment control and preserve green space in area slated for urban development. Prepared successful EPA-319 grant application for funding for project, assisted with detailed permitting efforts, coordinated efforts between private property owner and Laramie County, created design drawings for constructed wetland, assisted with public information efforts, prepared construction bid documents, conducted water quality monitoring, and provided construction management services.

Washington Avenue Phase 4 Final Design, City of Loveland, CO:

Responsible for project management. Project involved alternative analysis and final design for Phase 4 of Washington Avenue storm drainage system. Services included hydrologic and hydraulic analysis, storm drainage design, water line design and replacement, preparation of construction drawings and specifications, and coordination with multiple City departments. Significant utility crossings will include fiber optics, sanitary sewer, and sensitive aging water lines. Permits were obtained from railroads for jacking and boring storm sewer and water lines under three railroads. Water quality elements included four large water quality vaults called Carlsonators, which use extensive travel time to maximize settlement of smaller suspended solids. Three additional experimental water quality features were designed by Ayres Associates and City: bio-box (small bio-retention system); bio-basin (acts as filtration system); and modified storm inlet, which allows flows from smaller storm events to infiltrate into soil below inlet. Project was constructed in downtown Loveland in 2011. Construction of remaining portion of project was slated to begin in May 2012. Construction phase will have significant impact on large area of downtown. Meetings provided home and business owners with explanation of project.



Total Experience

13 Years

Registrations

*Registered Professional Engineer,
CO, 2007*

Certifications

*Class 1 Wastewater Collection
System Operator, CO, State of CO,
2010*

*Confined Space Training, Ayres
Associates, 2014*

Education

*BS, Civil Engineering, Colorado
State University, 2001*

Memberships

Water Environment Federation

Sam's training and experience are in the fields of sanitary sewer modeling and flow monitoring, storm drain design, and surface water hydraulics. His responsibilities include planning and overseeing flow monitoring projects, data analysis, hydraulic modeling, and scour analysis. He is proficient in MicroStation, HEC-RAS, SMS, SWMM, Flowlink, and H2Omap Sewer.

Here is what Matthew Fater of the City of Fort Collins Storm Water Utility had to say about Sam's work on the Fort Collins Inflow and Infiltration Reduction Study project: "Great job, Sam and Katie. This is not a glamorous job, but it's very important, so we appreciate your efforts."

Related Projects

Capitol Basin Storm Drain Assessment, City of Cheyenne: Project engineer. Project involved confined space entry into an 84-inch storm drain in City's historic downtown area to try to determine source of odors. Project team also collected water quality samples from storm drains upstream.

Lake Minnehaha Remediation Plan, City of Cheyenne: Project engineer. Project involved developing plan for ecological remediation of historic Lake Minnehaha in Holliday Park. Lake had high phosphorus and nitrogen levels that promote phytoplankton blooms, pH levels higher than most aquatic species can tolerate, water temperatures that exclude cold water fish, odors that indicate anoxic conditions in sediment, and limited depth. Tasks included water quality sampling, sediment sampling, locating site for disposal of dredged materials, environmental analysis, and preparation of remediation plan.

Remediation of Historic Lake Minnehaha Holliday Park – Phase 1, City of Cheyenne: Project engineer. **Remediation of Historic Lake Minnehaha Holliday Park – Phase 1, City of Cheyenne:** Responsible for project management. Project involved taking concepts and action points from remediation plan to full design. Items suggested in remediation plan were divided into Phase 1 tasks, which were primarily within boundary of Lake Minnehaha, and Phase 2 tasks, which were outside of lake. Phase 1 implemented deepening of lake, removal of island, upgrading lake's outlet structure, and outlet pipe and riprapping bank line.

Lake Minnehaha – Structural Design Work, City of Cheyenne: Project engineer. City of Cheyenne requested structural design work for retaining walls related to Lake Minnehaha Remediation. Ayres Associates' structural engineers in Eau Claire, Wisconsin, office worked with project engineers in Cheyenne and Fort Collins to respond quickly to structural design work and evaluate alternatives for cost comparison.

Crow Creek North Interceptor Sewer Rehabilitation, Cheyenne Board of Public Utilities, WY: Project engineer. Project involves design and construction services for rehabilitation of 7,000 feet of 30-inch to 36-inch concrete pipe sanitary sewer that has been offline for several years. Project challenges include groundwater infiltration, new flow diversion structure, 300 feet of new pipe, and installation of liner into existing pipe. Design effort includes reviewing closed circuit TV footage, confined space entry visual inspection, survey, potholing, groundwater review and modeling, new manholes, and coordination with private property owners.

Laramie County Lower Dry Creek Wetland, Laramie County Planning & Development Office: Project engineer. Project involved engineering services for 2.5-acre forebay pond with 2.5-acre wetland chase to provide sediment control and preserve green space in area slated for urban development. Prepared successful EPA-319 grant application for funding for project, assisted with detailed permitting efforts, coordinated efforts between private property owner and Laramie County, created design drawings for constructed wetland, assisted with public information efforts, prepared construction bid documents, conducted water quality monitoring, and provided construction management services.

Larimer County Bridge and Roadway NPS, LONCO, Inc., CO: Responsible for channel design. Project involved providing hydraulic design support of Larimer County Road 38E bridge replacement

over Redstone Creek including geomorphic assessment of channel, verification of effective hydrology, analysis of existing and proposed conditions bridge hydraulics, bridge scour analysis, bank stabilization and scour countermeasure design, and floodplain regulation compliance.

Washington Avenue Phase 4 Final Design, City of Loveland, CO: Project engineer. Project involved alternative analysis and final design for Phase 4 of Washington Avenue storm drainage system. Services included hydrologic and hydraulic analysis, storm drainage design, water line design and replacement, preparation of construction drawings and specifications, and coordination with multiple City departments. Significant utility crossings will include fiber optics, sanitary sewer, and sensitive aging water lines. Permits were obtained from railroads for jacking and boring storm sewer and water lines under three railroads. Water quality elements included four large water quality vaults called Carlsonators, which use extensive travel time to maximize settlement of smaller suspended solids. Three additional experimental water quality features were designed by Ayres Associates and City: bio-box (small bio-retention system); bio-basin (acts as filtration system); and modified storm inlet, which allows flows from smaller storm events to infiltrate into soil below inlet. Project was constructed in downtown Loveland in 2011. Construction of remaining portion of project was slated to begin in May 2012. Construction phase will have significant impact on large area of downtown. Meetings provided home and business owners with explanation of project.

Florian Fiebig, EIT

Water Resources Specialist



Total Experience

2 Years

Registrations

Engineer Intern, FL

Education

*MEng, Civil Engineering-Water
Resources and Hydraulics,
University of Florida, 2011*

*BS, Civil Engineering, University of
Florida, 2010*

Lake Minnehaha Remediation Plan Phase 2
City of Cheyenne

AYRES
ASSOCIATES

Florian joined Ayres Associates' water resources engineering staff in 2012, bringing experience in hydraulics, hydrology, two- and three-dimensional groundwater flow modeling, drainage and retention pond design, reactive contaminant transport modeling, and stormwater runoff monitoring. His experience includes monitoring wastewater effluent and investigating river system hydrology, biochemical dynamics, and nutrient levels.

Related Projects

Survey Monumentation, Photogrammetric Services, Major Drainageway Planning for the Strasburg Area, Adams County, CO:

Staff engineer. Ayres Associates is assisting Adams County with drainage and flood control improvements that will improve stormwater quality, provide floodplain management and preservation, address current and future erosion problems, and update hydrology for the watershed, among other benefits. Services include creating new topographic mapping, creating hydrologic and hydraulic models, evaluating Centennial Acres tributary flood hazard, developing major drainageway plan for Centennial Acres tributary, developing an outfall system plan for developed area of Strasburg, revising the Special Flood Hazard Area to include the Centennial Acres tributary, and facilitating formal public presentations.

Benson Park Pond and 29th Street Culvert Improvements, City of Loveland, CO:

Staff engineer. Project involves providing dredging and construction plans for removal of sediment from City of Loveland's Benson Park pond and installation of new, properly sized outlet pipe on 29th Street side. Challenges included protecting sculptures in and near pond, dense utility structures in 29th Street, and traffic volume on 29th Street. In addition, no construction plans existed for pond, requiring topographic survey.

Lewstone Creek-Scour Stabilization at Poudre Pipelines, City of Fort Collins, CO:

Staff engineer. Project involved designing erosion control and protective cover for City of Fort Collins drinking water pipelines where pipes cross Lewstone Creek. Damaging runoff had destabilized creek channel and created headcut from confluence with Poudre River upstream to the City's pipelines, exposing one pipe. Downstream from pipe crossing, Ayres designed partially grouted riprap drop structure. Existing wall was used as groundwater cutoff wall and was raised and extended laterally to address flanking issues. Due to prevailing groundwater levels, toe of partially grouted riprap was protected with launchable key of very large riprap that will arrest progress of headcutting when it reaches protected reach of creek. Upstream of cutoff wall, critical pipe crossing was stabilized and covered with structural concrete cap to maintain existing channel bottom elevation, which was only 6 inches above pipes.

Crow Creek North Interceptor Sewer Rehabilitation, Cheyenne Board of Public Utilities, WY:

Staff engineer. Project involves design and construction services for rehabilitation of 7,000 feet of 30-inch to 36-inch concrete pipe sanitary sewer that has been offline for several years. Project challenges include groundwater infiltration, new flow diversion structure, 300 feet of new pipe, and installation of liner into existing pipe. Design effort includes reviewing closed circuit TV footage, confined space entry visual inspection, survey, potholing, groundwater review and modeling, new manholes, and coordination with private property owners.

Pleasant Valley Pipeline Forebay Sedimentation Pond, City of Fort Collins, CO:

Staff engineer. Project involved adding sediment forebay to Pleasant Valley Pipeline diversion from Pleasant Valley canal. Work included civil/site design, structural design, geotechnical investigation and design, hydraulic and sediment analysis, project coordination, and design documentation. Project followed massive summer 2012 forest fire that brought potential of generating highly turbid runoff, possibly impacting City's Poudre River water supply. Ayres previously designed screening system to keep sediment, leaves, pine needles, and pine cones from clogging flow control valves at treatment plants, which used to require costly manual cleaning many times a day.

East Side Detention Facility, Pinnacle Consulting Group, Inc., CO:

Staff engineer. Project involves analyzing Gray Lakes Reservoir area as possible location for East Side Storage Facility as part of Boxelder Basin Regional Stormwater Master Plan. Amount of storage capacity will be analyzed to determine feasibility of Gray Lakes and/or another site upstream while accounting for irrigation storage needs. One option is using existing dam material to construct new dam farther west. Several options will be considered related to hydrologic benefits downstream. Phase 1 services include hydrographic survey and investigation of property ownership in and around proposed channel site between CR 50 and CR 52. Hazard classification for Gray Lakes Dam will be determined. Project will involve developing rough grading plan, cross sections, hydrologic model of proposed reservoir, updated

flow map, and cost estimate for Gray Lakes dam embankment.

Hydraulic Modeling and Floodplain Management, Sunstate Equipment Co.:

Staff engineer. Project involves providing hydraulic modeling and floodplain management consulting services to guide expansion of Sunstate Equipment building at 4228 Mulberry Street in Fort Collins, Colorado, in accordance with local floodplain regulations. Tasks include initial consultation with Larimer County Flood Review Board, data collection, hydraulic modeling analysis, assessment of hydraulic impacts, and as-built verification for building addition.

Emergency Water Line Protection, City of Loveland, CO:

Water resources specialist responsible for water and wastewater assessments. Two of three water lines feeding City of Loveland had been damaged in September 2013 floods, and remaining line was in danger of being undermined and damaged by raging Big Thompson River, which had changed course. Within four hours, Ayres team and City staff had conceptual plan to push river back to its original location. By following evening, river had been moved back under very difficult flood-flow conditions through earth-moving effort. Design and permitting processes normally would have taken years to complete. Ayres staff also joined Loveland's Emergency Operations Center (EOC), working all hours to troubleshoot problems, particularly with damaged utilities, and to begin navigating through Federal Emergency Management Agency (FEMA) process. Other staff assisted with geomorphic flood damage assessment and determined peak flow rate of event using high water measurements taken just after flood.

Major Drainage Plan for Big Dry Creek (Downstream of County Line Road), Urban Drainage and Flood Control District, CO:

Staff engineer. Project involves fieldwork and development and evaluation of all reasonable alternatives for drainage and flood control plans for Big Dry Creek downstream of County Line Road. Project will include stormwater modeling; planning for drainage, flood control, and stormwater quality; generation of alternatives for improvements; and the drafting of a final major drainageway conceptual design report.



Total Experience
2 Years

Registrations
Engineer-in-Training, CO, 2011

Education
*BS, Civil Engineering and Soil and
Water Engineering, Colorado
State University, 2012*

Pat joined Ayres in 2012. After graduating from CSU, Pat worked for the City of Fort Collins Utility Department. Since joining Ayres, Pat's expertise is in sanitary sewer flow monitoring, survey, construction administration, stormwater modeling, data analysis and drafting.

Related Projects

Dry Creek Master Plan EPA SWMM Conversion, City of Fort Collins, CO: Staff engineer. Project involves adding sediment forebay to Pleasant Valley Pipeline diversion from Pleasant Valley canal. Work includes civil/site design, structural design, geotechnical investigation and design, hydraulic and sediment analysis, project coordination, and design documentation. Project follows massive summer 2012 forest fire that brings potential of generating highly turbid runoff, possibly impacting City's Poudre River water supply.

Crow Creek North Interceptor Sewer Rehabilitation, Cheyenne Board of Public Utilities, WY: Staff engineer. Project involves design and construction services for rehabilitation of 7,000 feet of 30-inch to 36-inch concrete pipe sanitary sewer that has been offline for several years. Project challenges include groundwater infiltration, new flow diversion structure, 300 feet of new pipe, and installation of liner into existing pipe. Design effort includes reviewing closed circuit TV footage, confined space entry visual inspection, survey, potholing, groundwater review and modeling, new manholes, and coordination with private property owners.

Benson Park Pond and 29th Street Culvert Improvements, City of Loveland, CO: Staff engineer. Project involves providing dredging and construction plans for removal of sediment from City of Loveland's Benson Park pond and installation of new, properly sized outlet pipe on 29th Street side. Challenges included protecting sculptures in and near pond, dense utility structures in 29th Street, and traffic volume on 29th Street. In addition, no construction plans existed for pond, requiring topographic survey.

Lewstone Creek-Scour Stabilization at Poudre Pipelines, City of Fort Collins, CO: Staff engineer. Project involved designing erosion control and protective cover for City of Fort Collins drinking water pipelines where pipes cross Lewstone Creek. Damaging runoff had destabilized creek channel and created headcut from confluence with Poudre River upstream to the City's pipelines, exposing one pipe. Downstream from pipe crossing, Ayres designed partially grouted riprap drop structure. Existing wall was used as groundwater cutoff wall and was raised and extended laterally to address flanking issues. Due to prevailing groundwater levels, toe of partially grouted riprap was protected with launchable key of very large riprap that will arrest progress of headcutting when it reaches protected reach of creek. Upstream of cutoff wall, critical pipe crossing was stabilized and covered with structural concrete cap to maintain existing channel bottom elevation, which was only 6 inches above pipes.

Dell Range Corridor Study, Cheyenne

Metropolitan Planning Organization, WY: Staff engineer. Project involves analyzing intersections along Dell Range Boulevard corridor in commercial district of Cheyenne to identify solutions for safety, congestion, and access control concerns. Project involves many public comment meetings, business owner meeting, and public information meeting. Dell Range is major arterial roadway.

East Pershing Corridor Design and Construction, City of Cheyenne, WY:

Staff engineer. Project involved providing roadway and utility design and construction engineering for reconstruction of East Pershing Boulevard from Dunn Avenue to Converse Avenue, including reconstruction of East Pershing/Converse Avenue/19th Street intersection as five-leg, multi-lane roundabout. Project included public outreach, right-of-way survey, plan for phased construction to minimize traffic disruption, design of modular-block and cast-in-place concrete retaining walls in front of homes, and design of cantilevered sign structures directing traffic to correct lanes at roundabout. Also included were reconstruction of portions of Pershing Boulevard, replacement of water mains and associated valves and hydrants, replacement of six sanitary sewer manholes and 490 linear feet of sanitary sewer, installation of two new manholes, and relining or replacing additional 485 feet of sanitary sewer.

Pleasant Valley Pipeline Forebay Sedimentation Pond, City of Fort Collins, CO:

Staff engineer. Project involved adding sediment forebay to Pleasant Valley Pipeline diversion from Pleasant Valley canal. Work included civil/site design, structural design, geotechnical investigation and design, hydraulic and sediment analysis, project coordination, and design documentation. Project followed massive summer 2012 forest fire that

brought potential of generating highly turbid runoff, possibly impacting City's Poudre River water supply. Ayres previously designed screening system to keep sediment, leaves, pine needles, and pine cones from clogging flow control valves at treatment plants, which used to require costly manual cleaning many times a day.

Iowa State University Flood Mitigation, Iowa State University:

Staff engineer. Project involved developing strategy for flood mitigation for three facilities on Iowa State University campus. Several variables were considered in process of evaluating flood mitigation alternatives. Recommended solutions and provided cost estimates. Subconsultant (Shive-Hattery) provided summary of structural assessment of each building site.

Emergency Water Line Protection, City of Loveland, CO:

Water resources specialist responsible for water and wastewater assessments. Two of three water lines feeding City of Loveland had been damaged in September 2013 floods, and remaining line was in danger of being undermined and damaged by raging Big Thompson River, which had changed course. Within four hours, Ayres team and City staff had conceptual plan to push river back to its original location. By following evening, river had been moved back under very difficult flood-flow conditions through earth-moving effort. Design and permitting processes normally would have taken years to complete. Ayres staff also joined Loveland's Emergency Operations Center (EOC), working all hours to troubleshoot problems, particularly with damaged utilities, and to begin navigating through Federal Emergency Management Agency (FEMA) process. Other staff assisted with geomorphic flood damage assessment and determined peak flow rate of event using high water measurements taken just after flood.

**Andrea Faucett,
PE, CFM**

Senior Project Manager



Total Experience

31 Years

Registrations

*Registered Professional Engineer,
CO, 1990; IA, 2012*

Certifications

Certified Floodplain Manager, 2006

Education

*BS, Civil Engineering, Colorado
State University, 1983*

Andrea has more than 31 years of experience in water resources and municipal engineering covering a wide range of design, including hydrology and hydraulics, open-channel hydraulics, roadway drainage, storm sewer hydraulics, water quality, detention and sediment basins, dam design, stream stabilization, permitting, floodplain analysis and mapping, CLOMR/LOMR preparation, wetland restoration, and utility design. She has performed as project manager/project engineer on numerous projects, completing or overseeing the design from the analysis phase through construction plan and specification preparation.

Chris Carlson, a civil engineer with the City of Loveland, Colorado, had this to say about Andrea's work on the Washington Avenue Phase 4 Storm Drainage Design project: "Andrea is an outstanding project manager and always provides a good teamwork atmosphere for everyone involved with the project. When there were significant difficulties with the project's direction during design, Andrea stepped in to save the day. We are very grateful."

Related Projects

Canal Importation Pond and Outfall Design (New Mercer Ditch), City of Fort Collins, CO: Project manager responsible for project administration and design oversight. Largest stormwater project undertaken by City of Fort Collins since 1997 flood involves design and construction of \$20 million of stormwater improvements in heart of City directly upstream of Colorado State University (CSU). Extensive public process including mailings, open houses, and project website to educate public on project's impact on neighborhood roadways and primary traffic corridors serving CSU. Project involved reducing spills from three drainage ditches that currently impact more than 175 homes during 100-year storm event. Project included complex hydrologic and hydraulic modeling (utilizing SWMM and unsteady HEC-RAS) to determine ditch spill quantities and mapping of existing floodplain. Project included design of five regional detention/water quality ponds and more than 4,000 feet of large diameter (up to 102 inches) storm sewer and associated laterals and inlets. Inlets include use of Snouts and upstream "sweeper sections" to improve water quality in downstream receiving waters. Project included design of round version of Bureau of Reclamation Type 10 hanging baffle structure.

Lake Minnehaha Remediation Plan, City of Cheyenne, WY: Technical expert. Project involved developing plan for ecological remediation of historic Lake Minnehaha in Holliday Park. Lake has high phosphorus and nitrogen levels that promote phytoplankton blooms, pH levels higher than most aquatic species can tolerate, water temperatures that exclude cold water fish, odors that indicate anoxic conditions in sediment, and limited depth. Tasks included water quality sampling, sediment sampling, locating site for disposal of dredged materials, environmental analysis, and preparation of remediation plan.

Benson Park Pond and 29th Street Culvert Improvements, City of Loveland, CO: Responsible for project management. Project involves providing dredging and construction plans for removal of sediment from City of Loveland's Benson Park pond and installation of new, properly sized outlet pipe on 29th Street side. Challenges included protecting sculptures in and near pond, dense utility structures in 29th Street, and traffic volume on 29th Street. In addition, no construction plans existed for pond, requiring topographic survey.

Dry Creek Sheridan Reach Flood Control Improvement, City of Cheyenne, WY: Managed hydrologic and hydraulic modeling. Responsible for preparing construction plans and specifications. Designed \$5.5 million Dry Creek Flood Control Improvement project to address City of Cheyenne's most hazardous drainage reach. Existing condition hydrology was prepared with HEC-1 model that was updated to reflect improvements. Flood control project involved designing diversion of all but 5-year storm flows from main channel of Dry Creek into three detention ponds before discharging attenuated flows back into Dry Creek downstream. Innovative final design and successful negotiations with affected property owners led to construction cost savings of approximately \$1.4 million. Project included design of major pedestrian underpass under arterial road and constructed wetland to control sediment and other pollutants.

Dry Creek Master Plan EPA SWMM Conversion, City of Fort Collins, CO: Responsible for project management. Project involves adding sediment forebay to Pleasant Valley Pipeline diversion from Pleasant Valley canal. Work includes civil/site design, structural design, geotechnical investigation and design, hydraulic and sediment analysis, project coordination, and design documentation. Project follows massive summer 2012 forest fire that brings potential of generating highly turbid runoff, possibly impacting City's Poudre River water supply.

Clark Reservoir Boat Dock Pier, Larimer County, CO: Responsible for project management. Project involves provide hydrologic and hydraulic analysis, engineering design, surveying, plan preparation, and construction management services on non-project-

specific work order basis for three flood control projects.

Coal Creek Diversion and Clark Reservoir Storage, Larimer County, CO: Responsible for project management. Project involved providing hydrologic and hydraulic analysis, engineering design, surveying, plan preparation, and construction management services on diversion of Coal Creek. Project is part of Boxelder Basin Flood Control project that involves providing hydrologic and hydraulic analysis, engineering design, surveying, plan preparation, and construction management services on work-order basis for three flood control projects on Boxelder Creek.

Lewstone Creek-Scour Stabilization at Poudre Pipelines, City of Fort Collins, CO: Assistant project manager. Project involved designing erosion control and protective cover for City of Fort Collins drinking water pipelines where pipes cross Lewstone Creek. Damaging runoff had destabilized creek channel and created headcut from confluence with Poudre River upstream to the City's pipelines, exposing one pipe. Downstream from pipe crossing, Ayres designed partially grouted riprap drop structure. Existing wall was used as groundwater cutoff wall and was raised and extended laterally to address flanking issues. Due to prevailing groundwater levels, toe of partially grouted riprap was protected with launchable key of very large riprap that will arrest progress of headcutting when it reaches protected reach of creek. Upstream of cutoff wall, critical pipe crossing was stabilized and covered with structural concrete cap to maintain existing channel bottom elevation, which was only 6 inches above pipes.

East Vine Drive Realignment, City of Fort Collins, CO: Responsible for project management. Project involved completing design plans for relocation of East Vine Drive. First phase involved preliminary design for relocated East Vine Drive between North College Avenue and Dry Creek, consisting of design for new roadway approximately 1.3 miles long. Second phase involved conceptual design for continuation of relocated East Vine Drive from Dry Creek to connection back to existing East Vine Drive.

Darci Hendon, PE

Civil Engineer



Total Experience

15 Years

Registrations

Registered Professional Engineer,
WY; 2005; CO, 2006

Education

BS, Civil Engineering, University of
Wyoming, 1999

Memberships

National Society of Professional
Engineers

Darci joined Ayres Associates in 2006. She brings 15 years of civil and transportation engineering experience with additional experience on subdivision and site design. She is proficient using AutoCAD Civil 3D, MicroStation, GEOPAK, and Hasestad Methods software (StormCAD, WaterCAD, and SewerCAD).

Related Projects

Carey Avenue Road Reconstruction, City of Cheyenne, WY: Project engineer. Project involved design for extension of Carey Avenue in downtown area of Cheyenne. Services included engineering design and technical assistance during construction. Project involved coordination with utilities and with City's urban planning office. One goal of project design was to construct model corridor for use as example of what urban planning elements could be included in downtown road reconstruction project.

Cheyenne Greenway Extension Phase III, City of Cheyenne, WY: Project engineer. Provided review of design and bid documents for City of Cheyenne's Phase III Greenway pedestrian pathway project. Project involved coordinating with Union Pacific Railroad to obtain easement on railroad property for placement of pathway, providing City substantial cost savings over initial alignment. Project includes 8,100 feet of pedestrian pathway and 75-foot-long pedestrian bridge. Ayres Associates oversaw design of pedestrian bridge and abutments.

Converse Ball Field Improvements and Parking Lot, DHM Designs for City of Cheyenne: Assistant project manager. Project involves improvements to existing four-field sports complex in Cheyenne. One field will be made into "championship" field, and all fields will have improved lighting, fence, and in-field mix. Ayres is preparing site plan review documents, designing new parking lot, and designing drainage improvements.

East Pershing Corridor Design and Construction, City of Cheyenne, WY: Project engineer. Project involved providing roadway and utility design and construction engineering for reconstruction of East Pershing Boulevard from Dunn Avenue to Converse Avenue, including reconstruction of East Pershing/Converse Avenue/19th Street intersection as five-leg, multi-lane roundabout. Project included public outreach, right-of-way survey, plan for phased construction to minimize traffic disruption, design of modular-block and cast-in-place concrete retaining walls in front of homes, and design of cantilevered sign structures directing traffic to correct lanes at roundabout. Also included were reconstruction of portions of Pershing Boulevard, replacement of water mains and associated valves and hydrants, replacement of six sanitary sewer manholes and 490 linear feet of sanitary sewer, installation of two new manholes, and relining or replacing additional 485 feet of sanitary sewer.

Holliday Park Connector, City of Cheyenne,

WY: Project manager. Project involved design of greenway path that connects existing greenway facilities to Holliday Park. Project began with analysis of options for grade-separated crossing of Lincolnway. Stakeholders were invited to provide comments on options, with underground crossing selected. Underpass is 8-foot-high, 12-foot-wide precast concrete box. Design replaced free-flow right-turn lane at Lincolnway with dedicated right-turn lane; realigned intersection of 13th Street/Dunn Avenue and Nationway; and included new storm sewer facilities and improvements to water and sanitary sewer facilities.

Arp Elementary School Greenway Connector,

City of Cheyenne, WY: Project engineer. Project involved producing final plans and construction documents for nearly 1-mile section of greenway path in South Cheyenne to connect two subdivisions to Arp Elementary School. Project also involved relocating a fire hydrant and designing a new drainage ditch.

East Vine Drive Realignment, City of Fort

Collins, CO: Project engineer. Project involved completing design plans for relocation of East Vine Drive. First phase involved preliminary design for relocated East Vine Drive between North College Avenue and Dry Creek, consisting of design for new roadway approximately 1.3 miles long. Second phase involved conceptual design for continuation of relocated East Vine Drive from Dry Creek to connection back to existing East Vine Drive. Design included location and potholing of existing utilities within project corridor, geotechnical investigation and pavement design by subconsultant, and preparing for and attending public outreach meetings, and Fort Collins City Council meetings. Design included street drainage design, cross sections, and horizontal and vertical alignment of road.

Laramie County School District No. 1 – Repairs and Replacement of Parking Lots and Drives 2015, Laramie County School District No. 1, WY:

Project manager. Ayres is completing construction plans and performing construction inspection for

selected Laramie County School District sites: McCormick Junior High School, Pioneer Park Elementary School, and East High School. Work at McCormick includes converting an asphalt fire lane to a concrete pavement; replacing curb, gutter, sidewalk, and asphalt pavement in a student drop off lane; converting roof drains along the student drop off lane to an underground collection system which will be piped to a detention pond. The Pioneer Park Elementary site includes complete reconstruction of their main parking lot to address surface drainage concerns. The East High School site includes a mill and overlay design, with sections of new curb and gutter for one of their minor parking lots.

Laramie County School District No. 1: Fire Lane Project, Laramie County School District No. 1,

WY: Project manager. Project involved design and construction administration for replacing fire lane at Central High School, along with miscellaneous concrete and asphalt work. Design addressed drainage issues that caused erosion on slope and damage to retaining wall. Fire lane was paved with concrete pavement, and select locations of curb and gutter and sidewalk were removed and replaced. Complete storm sewer system was installed, including pipe, manholes, inlets, and an outfall to a constructed swale. Design included placement of concrete pad for bleachers at football field and providing Americans with Disabilities Act-accessible route to access new bleachers.

**Remediation of Historic Lake Minnehaha
Holliday Park – Phase 1, City of Cheyenne:**

Project engineer. Project involved taking concepts and action points from remediation plan to full design. Items suggested in remediation plan were divided into Phase 1 tasks, which were primarily within boundary of Lake Minnehaha, and Phase 2 tasks, which were outside of lake. Phase 1 implemented deepening of lake, removal of island, upgrading lake's outlet structure, and outlet pipe and riprapping bank line.

Samantha Campbell, EIT

Transportation/Traffic Engineering
Staff



Total Experience

4 Years

Registrations

Engineer-in-Training, ID, 2010

Education

MS, Civil Engineering, University of
Idaho, 2011

BS, Civil Engineering, University of
Idaho, 2010

Samantha joined Ayres Associates in 2012. She brings experience in transportation and construction analysis and design, permitting, surveying, and construction inspection. She has also provided analysis of traffic counts, crash data, and future forecasts, and she has produced traffic flow models using Synchro. Samantha's master's thesis included researching and compiling a curriculum to teach students and professionals the process of railroad preemption.

Related Projects

Central High School Riske Field Renovations, Laramie County

School District No. 1: Project engineer. Ayres was retained to complete construction drawings for the new grandstands and the Riske Field site. The final design of the site includes a grandstand system supported by a drilled pier foundation, seating for 1,504 spectators, a new building which will include restrooms, concessions, a ticket window, and an IT room. Additional site improvements include new sidewalk, emergency access, pedestrian lighting, scoreboard, and improvements to the existing storm sewer system. Construction is underway and expected to be complete by August 2015.

Cheyenne HGI Hotel Project, Pahl Architecture:

Project engineer. Ayres provided engineering services related to the 10-story Hilton Garden Inn hotel in downtown Cheyenne, Wyoming. The hotel site includes a sky bridge to the east providing access to an existing parking garage on a highly visible corner. Services include grading design, drainage design, ADA compliance, striping plan, signage plan, and all utility design (water, sanitary sewer, storm sewer, gas, cable, telephone, electric). Several existing utilities had to be relocated for the design of the hotel facilities.

East Pershing Corridor Design and Construction, City of Cheyenne:

Responsible for construction inspection and final plan preparation. Project involved providing roadway and utility design and construction engineering for reconstruction of East Pershing Boulevard from Dunn Avenue to Converse Avenue, including reconstruction of East Pershing/Converse Avenue/19th Street intersection as five-leg, multi-lane roundabout. Project included public outreach, right-of-way survey, plan for phased construction to minimize traffic disruption, design of modular-block and cast-in-place concrete retaining walls in front of homes, and design of cantilevered sign structures directing traffic to correct lanes at roundabout. Also included were reconstruction of portions of Pershing Boulevard, replacement of water mains and associated valves and hydrants, replacement of six sanitary sewer manholes and 490 linear feet of sanitary sewer, installation of two new manholes, and relining or replacing additional 485 feet of sanitary sewer.

Holliday Park Connector, City of Cheyenne:

Responsible for construction observation. Project involved design of greenway path that connects existing greenway facilities to Holliday Park. Project began with analysis of options for grade-separated crossing of Lincolnway. Stakeholders were invited to provide comments on options, with underground crossing selected. Underpass is 8-foot-high, 12-foot-wide precast concrete box. Design replaced free-flow right-turn lane at Lincolnway with dedicated right-turn lane; realigned intersection of 13th Street/Dunn Avenue and Nationway; and included new storm sewer facilities and improvements to water and sanitary sewer facilities.

Repairs and Replacement of Parking Lots and Drives, Laramie County School District No. 1:

Staff engineer. Ayres is completing construction plans and performing construction inspection for selected Laramie County School District sites: McCormick Junior High School, Pioneer Park Elementary School, and East High School. Work at McCormick includes converting an asphalt fire lane to a concrete pavement; replacing curb, gutter, sidewalk, and asphalt pavement in a student drop off

lane; converting roof drains along the student drop off lane to an underground collection system which will be piped to a detention pond. The Pioneer Park Elementary site includes complete reconstruction of their main parking lot to address surface drainage concerns. The East High School site includes a mill and overlay design, with sections of new curb and gutter for one of their minor parking lots.

Lena Gulch at South Golden Road Final

Design, Jefferson County, CO: Responsible for review of traffic control plan. Project involves final design plans and specifications for drainage and flood control improvements for Lena Gulch at South Golden Road. Improvements include drop structure upstream of South Golden Road; concrete box culvert for Lena Gulch under road; channel improvements downstream of road; roadway reconstruction for 1,000 feet of South Golden Road; and storm sewer design for roadway improvements that will accommodate existing major fiber optic cables, medium pressure gas pipe, sanitary sewers, and water distribution pipes within street right-of-way.

JEFFREY B. JONES
STATEMENT OF QUALIFICATIONS



Education:

High School: Torrington High School
Torrington, Wyoming, 1980

College: Northwest Community College
Powell, Wyoming
A. S. Forestry, 1982

Oregon State University
Corvallis, Oregon
B.S. Forest Engineering, 1985

License:

Engineering Training, State of Oregon
E.I.T. #7119, 1985

Land Surveyor In Training, State of Oregon
L.S.I.T. #874, 1985

Professional Land Surveyor, State of Wyoming
P.L.S. #5910, 1990

Professional Land Surveyor, State of Colorado
P.L.S. #37920, 2004

Professional Work Experience:

1986-1994-Intermountain Professional Services:
Party Chief and Principal responsible for survey department.
Extensive experience in subdivisions, ALTA Surveys, Aerial photo
control, construction staking and water rights.

1994-Present-Steil Surveying Services, LLC:
Professional Land Surveyor in charge of several field crews.
Extensive experience in GPS control networks, GPS error analysis,
use and limitations of GPS equipment, subdivisions, construction
staking, ALTA surveys, water rights, cadastral surveys and
research.

BRENT F. WILKINS, P.E.

GEOTECHNICAL DEPARTMENT MANAGER

PROFESSIONAL EXPERIENCE

Mr. Wilkins is the manager of the geotechnical engineering department in Terracon's Cheyenne, Wyoming office. He has over 25 years experience in geotechnical and civil engineering and construction quality management. He supervises the geotechnical operations of the office and provides engineering consultation on projects including foundations, roadways and construction quality control and materials testing.

Mr. Wilkins' primary experience is in subsurface exploration and geotechnical engineering evaluation for building and bridge foundations, roadways, landfills and foundation forensic investigations. He provides engineering analysis and recommendations for foundation settlement, soil bearing capacity, soil lateral resistance, liquefaction potential and soil seismic parameters. He is experienced in the evaluation of shallow and deep foundations, slope stability and ground water control. He has prepared engineered designs and construction specifications for retaining walls, stabilized slopes, large pipe-arch culvert foundations and landfills.

Mr. Wilkins' experience also includes the management of geotechnical exploration activities and laboratory soil testing, engineering evaluation and design of pipe-arch structures, extensive construction QA/QC monitoring, landfill design and permitting and surface mine permitting.

PROJECT EXPERIENCE

Fiscal Years 2006 – 2012 Water and Sewer Rehabs – Cheyenne Board of Public Utilities

Project manager for the geotechnical exploration for the reconstruction of yearly water and sewer improvements throughout Cheyenne.

Pioneer & Carey Avenues Reconstruction – City of Cheyenne

Project manager and senior engineer for the geotechnical exploration, testing and analysis of the reconstruction of eleven blocks of roadway and utility improvements. In addition to standard subgrade profiles, existing pavement thickness and condition profiles were provided at multiple sections within each block. Pavement design recommendations were provided for alternative combinations of rehabilitation and reconstruction.

Waste Water Treatment Plants – City of Cheyenne

Project manager for geotechnical exploration and laboratory testing for substantial remediation and additions to the Dry Creek and Crow Creek wastewater treatment plants. Provided engineering design recommendations for shallow and deep spread footing foundations, underground basins, lateral earth pressures, dewatering, trench stability, and construction issues.

30" Water Main Extension – Board of Public Utilities

Project manager for the geotechnical exploration and site evaluation for the extension of the Western Hills main to the Round Top Water Treatment Plant.

Sewage Lift Station – City of Cheyenne

Project manager for the geotechnical exploration and site evaluation for North Park Soccer Complex sewer extension and lift station. Provided spread footing foundation recommendations for the lift station and trenching/backfill recommendations for the sewer extension.

Education

*Master of Science,
Civil Engineering,
South Dakota School of Mines and
Technology, 1995*

*Bachelor of Science,
Civil Engineering,
University of Missouri-Rolla, 1986*

Registrations

*Professional Engineer:
Wyoming, No.9459*

Affiliations

*National Society of Professional
Engineers (NSPE)*

American Society of Civil Engineers

Work History

*Terracon Consultants, Inc.,
Department Manager,
2001-present*

*Bowser-Morner Associates, Inc.,
Senior Engineer,
1996-2001*

*Golder Associates, Inc.,
Staff Engineer,
1989-1994*

*Crockett Engineering Consultants,
Staff Engineer,
1987-1989*

Terracon

Schedule

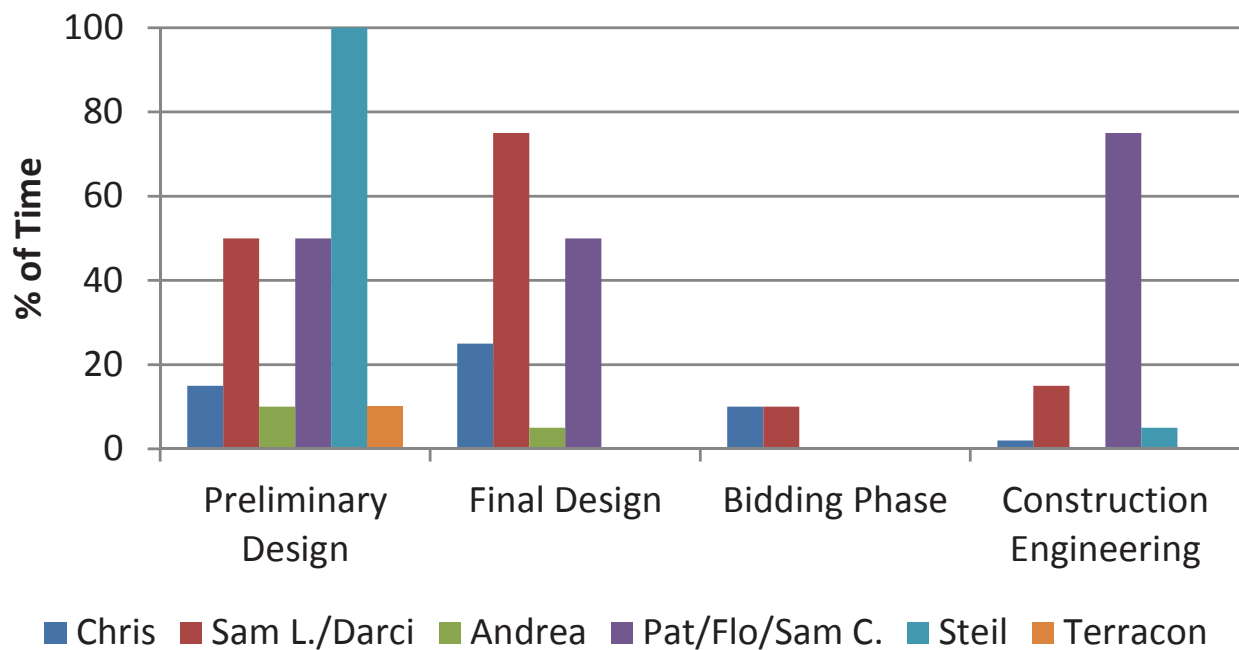
Ayres Associates carefully went through each step in the design and bidding process based on the outline in the City of Cheyenne's Request for Proposals. No matter how we dissect it, the start of construction can be no earlier than December 2015; more realistically, the start of construction will be in early January 2016. **A detailed schedule is included on the following page.**

Our hope was to be able to start construction immediately after Frontier Days to allow for three to four months of nice weather during construction. However, this does not seem possible. As currently

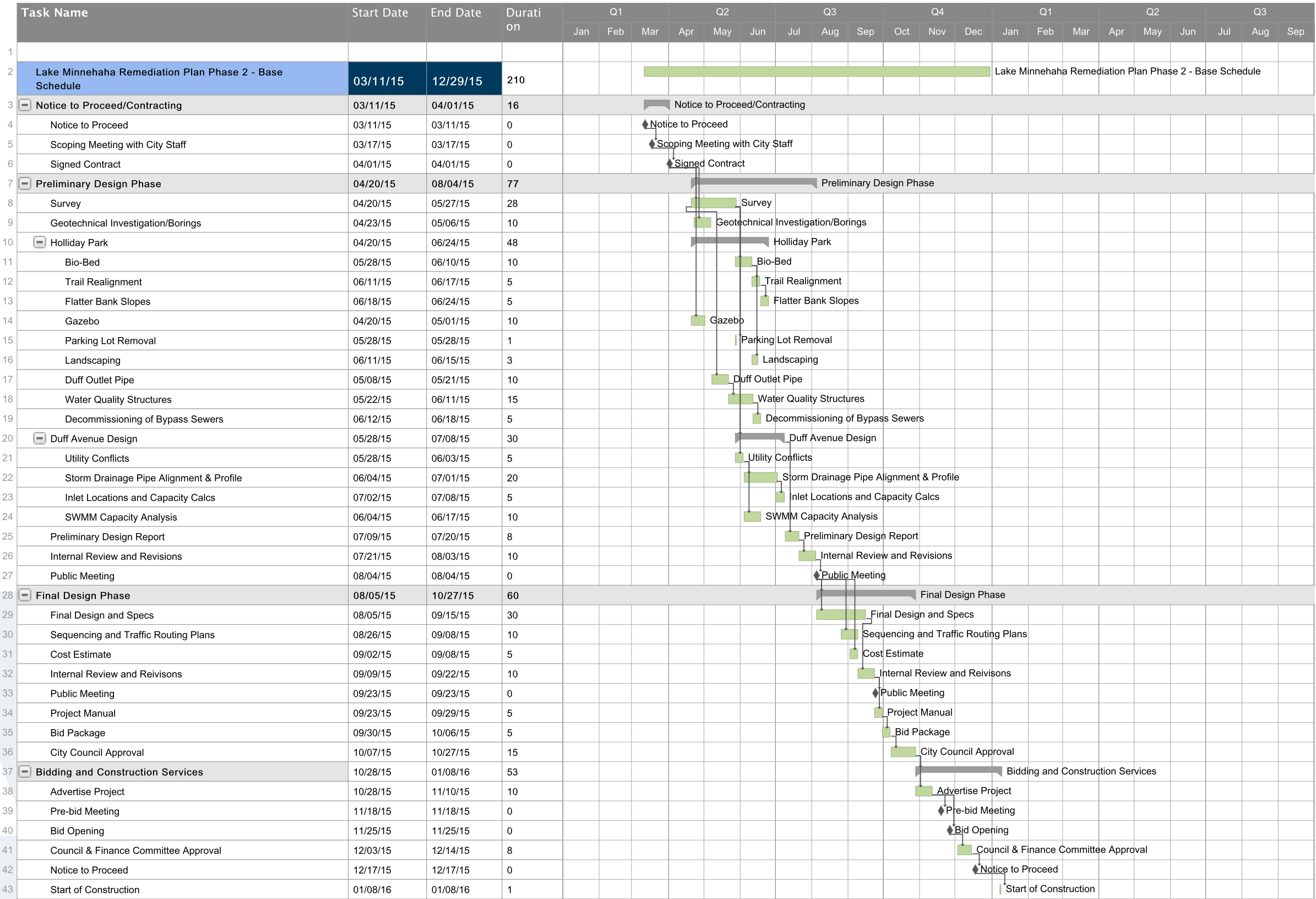
scheduled, a contractor can be selected in December 2015 with a construction start date slated for no later than April 2016. This will allow for lower bids, a better construction window, and a cleaner completion for the project – allowing for seeding, restoration, and complete project cleanup before Frontier Days 2016.

Ayres has the staff availability to commit several engineers to the design from the Notice to Proceed and will maintain the same committed staff throughout the project. **Following is a chart showing our staff availability.**

Project Team Availability



Project Schedule



Project Area



The Ayres project team is very familiar with the project area, having worked on a number of projects within Holliday Park. Ayres has worked closely with the Parks Department on the previous Lake Minnehaha Remediation Plan as well as the Phase 1 design and construction. During the Phase 1 design process, Ayres and the Parks Department discussed at length most items included in the Phase 2 project.

During the course of the Phase 1 design, Ayres investigated the storm sewer lines in the park to come up with preliminary water quality designs on the system upstream of the lake. This involved surveying and dipping inverts of the existing storm sewer pipes, particularly in the vicinities of Morrie Avenue, the intersection of Bradley Avenue and 19th Street, and the bypass lines north of the lake.

Our previous work on Lake Minnehaha uniquely positions us to be ahead of schedule on the preliminary design.

Additionally, Ayres staff designed the Holliday Park Connector, providing a pedestrian trail underneath Lincolnway into Holliday Park.



Scope of Work

After reading through the project's scope of work, Ayres Associates wholeheartedly agrees with each aspect of the project and understands how each item will enhance the water quality in Lake Minnehaha and the beauty of Holliday Park while simultaneously reducing flood damage within the drainage basin.

Routing stormwater from the Duff Avenue line through Lake Minnehaha will increase water circulation in the lake. The addition of water quality best management practices (BMPs) to the storm sewer system upstream of the lake will keep trash, oil, and sediment from polluting the lake and becoming an eyesore.

Ayres has designed multiple storm sewer projects within street corridors and will use our expertise in preparing the preliminary design of the storm sewer in Duff Avenue. The repairing of Duff Avenue after laying the storm sewer will also be closely considered. Due to the narrow nature of Duff Avenue, it is anticipated that the entire street section will be replaced (curb to curb) along most of the alignment.

The preliminary design of the storm sewer from Lake Minnehaha, up Duff Avenue to the box culvert underneath Pershing Boulevard, is nearly a half-mile long. Identifying and solving utility conflicts will be the major challenge of the preliminary design. Ayres anticipates that waterline lowerings, relocating sanitary sewer lines, and moving sanitary sewer drop manholes will solve most of the major conflicts. Additionally, Ayres will consider multiple barrels and elliptical pipe to avoid conflicts and to keep minimum cover. When the major conflicts are recognized early in the preliminary design phase, Ayres will order utility potholing to determine exact depths of the conflicting utilities to ensure a robust preliminary design as a final product.

The topography from Pershing Boulevard generally drops down to 19th Street; however, there is a high point in the road at 22nd Street. The topographical survey along with initial profile drawings of the proposed storm sewer will reveal the shallow cover north of 22nd Street and depth of pipe at the high point at 22nd Street. Ayres is already thinking of ideas to minimize the impact of deep trenching in that vicinity of the preliminary design.



Previous storm sewers Ayres has designed have incorporated multiple different types of stormwater BMPs, including proprietary, non-proprietary, bio-engineered beds, filtration inlets, water quality vaults, and water quality manholes. Additionally, we have kept in contact with our clients who are responsible for maintaining the BMPs, and they have provided a wealth of knowledge regarding the pros and cons of the various devices. We look forward to sharing with the City our findings and opinion from these installations.

Much of the Parks Department's planned improvements have already been taken to a preliminary level by our team under previous contracts and would be ready for review with minimal effort. Most notably the gazebo, trail replacement and relocation, bio-bed, stormwater BMPs, sprinkler repair, and construction have all been thoroughly discussed with the Parks Department, and in most cases have been taken to preliminary design.

At the west end of the lake, Ayres looks forward to designing features that will enhance the park, most notably the flattened bank slopes. Where the 36-inch and 42-inch storm sewers outfall into the lake, the flattened bank will provide functional aesthetics to the park where currently there is only a failing rock-embedded concrete slope. The function will come from an engineered boulder drop structure guiding the overflow channel's water to the lake level. The aesthetics will come from riparian plantings that thrive along a shoreline fringe.

In other areas along the north bank of the lake, the banks are steep and the trail is right along the top of

the bank. In these areas the asphalt trail is crumbling as the bank erodes beneath it. In these locations a combination of flattening the bank and relocating the trail will provide safety to pedestrians and reduce park maintenance.

Valuable assets that Ayres brings to this project include:

- Traffic engineers
- Storm drainage engineers
- Extensive SWMM modeling experience
- Hosting public meetings
- Landscape architects
- Extensive familiarity with Lake Minnehaha and Holliday Park

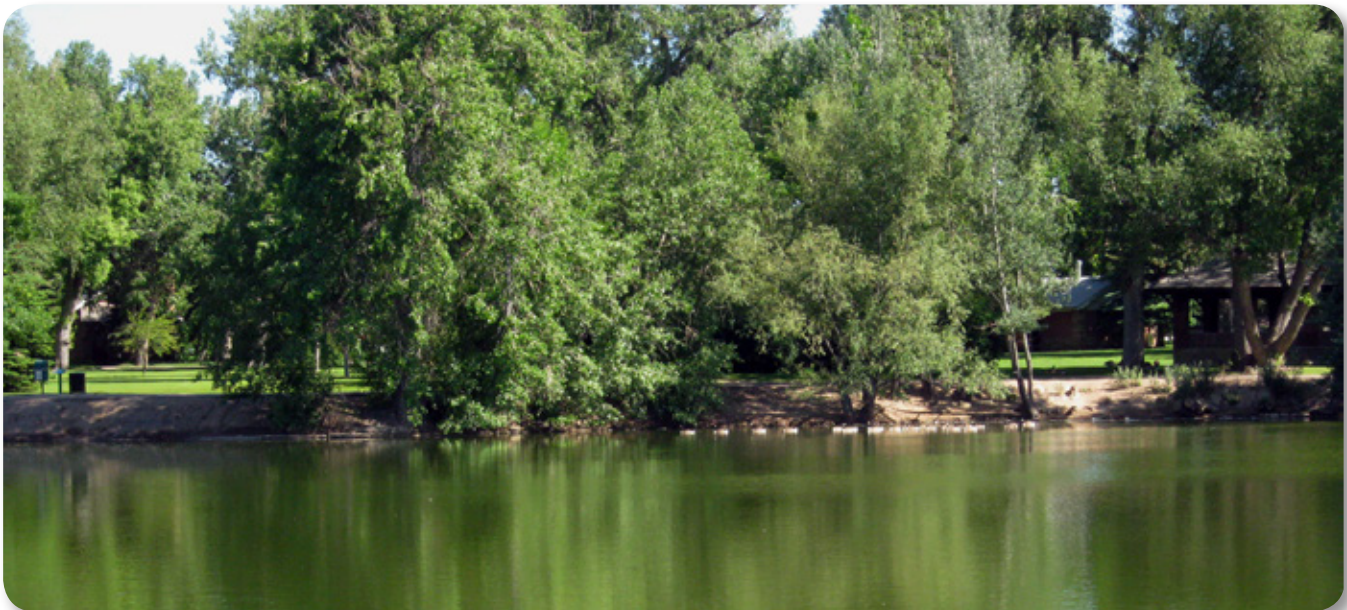
Ayres is honored to have been selected as the design engineer for the original Lake Minnehaha Remediation Plan and the Phase 1 design that was constructed last winter. Phase 1 corrected most of the problems associated with Lake Minnehaha, and last summer the lake looked better than it has in more than 30 years. Phase 2 improvements will ensure the lake remains pristine by eliminating two of the causes of poor lake health. First, routing the stormwater from the Duff Avenue storm sewer into the lake will increase the circulation occurring within the lake. Secondly, by providing water quality treatment to all storm sewers discharging into the lake sediment, oils and trash will no longer pollute Lake Minnehaha. The Phase 2 improvements will be the final step to keeping Lake Minnehaha a beautiful lake in the heart of historic Holliday Park, and Ayres would be honored to be able to see this project through from beginning to end.

References

Ayres Associates has completed several similar projects for the City of Cheyenne and other communities, including Phase 1 of the Lake Minnehaha Remediation Plan. Following are project examples that demonstrate our expertise for providing services on your project.

The best indication of our performance is to talk to people who use Ayres Associates' services. We encourage you to contact the references listed with each project for a firsthand account. We are confident you will get a fair picture of our method of conducting business.

City of Cheyenne Lake Minnehaha Remediation Plan



Client Name: City of Cheyenne
Address: 2101 O'Neil Avenue, Cheyenne, WY 82001
Contact Person: Sam Berta
Telephone: 307.637.6200
Project Dates: March 24, 2011 - January 16, 2012
Original Project Budget: \$39,683
Final Project Cost: \$39,683
Explanation of Variation from Original Budget to Final Project Cost: Completed on budget

The City of Cheyenne retained Ayres Associates to develop a plan for ecological remediation of historic Lake Minnehaha in Holliday Park. The lake has high phosphorus and nitrogen levels that promote eutrophic conditions such as phytoplankton blooms, pH levels higher than most aquatic species can tolerate, water temperatures that exclude cold water fish, odors that indicate anoxic conditions in the sediment, and limited depth. The remediation plan addresses each of these issues. Tasks included water quality sampling, sediment sampling, locating a site for disposal of dredged materials, environmental analysis, and preparation of the remediation plan.

Remediation of Historic Lake Minnehaha Holliday Park– Phase 1



Client Name: City of Cheyenne
Address: 2101 O'Neil Avenue,
Cheyenne, WY 82001
Contact Person: Rick Parish
Telephone: 307.638.4357
Project Dates: September 12,
2012 - August 2, 2013
Original Project Budget:
\$83,580
Final Project Cost: N/A
**Explanation of Variation
from Original Budget to Final
Project Cost:** N/A

The City of Cheyenne retained Ayres Associates to take the concepts from the remediation plan from action points to full design. The items suggested in the remediation plan were divided into Phase 1 tasks, which were primarily within the boundary of Lake Minnehaha, and Phase 2 tasks, which were outside of the lake.

Phase 1 implemented the deepening of the lake, removal of the island, upgrading the lake's outlet structure, and outlet pipe and riprapping the bank line.

East Pershing Corridor Design and Construction



The City of Cheyenne retained Ayres Associates to provide roadway and utility design and construction engineering for the reconstruction of East Pershing Boulevard from Dunn Avenue to Converse Avenue, including reconstruction of the East Pershing/Converse Avenue/19th Street intersection as a five-leg, multi-lane roundabout. The project included providing extensive public outreach, right-of-way survey, a plan for phased construction to minimize traffic disruption, design of modular-block and cast-in-place concrete retaining walls in front of homes, and the design of cantilevered sign structures directing traffic to the correct lanes at the roundabout.

Client Name: City of Cheyenne
Address: 2101 O'Neil Avenue, Cheyenne, WY 82001
Contact Person: Nathan Beauheim
Telephone: 307.638.4315
Project Dates: January 4, 2010 - December 1, 2013
Original Project Budget: \$1,319,946
Final Project Cost: \$1,319,946
Explanation of Variation from Original Budget to Final Project Cost: Completed on budget

Complete reconstruction was required on Pershing Boulevard from Logan Avenue to Converse Avenue. From Dunn Avenue to Logan Avenue, rehabilitation of the existing section of Pershing was selected as an alternative to total reconstruction. The signalized intersection of Pershing and Concord Road/Logan Avenue was examined during the design phase. Improvements were made to the existing traffic signal system at both cross streets to better accommodate traffic flow through the intersections.

The existing 6-inch and 12-inch parallel water mains were replaced and combined into one main, and associated valves and hydrants were replaced. Special attention was paid to facilitating future utility maintenance at the roundabout with minimal disruption to traffic. Six sanitary sewer manholes and 490 linear feet of sanitary sewer were replaced, along with adding two new manholes and relining or replacing an additional 485 feet of sanitary sewer.

Washington Avenue Phase 4 Final Design



The City of Loveland retained Ayres Associates for a five-year, open-ended contract. The first project under the contract involved an alternative analysis and final design for Phase 4 of the Washington Avenue storm drainage system.

Services for this project included hydrologic and hydraulic analysis, storm drainage design, water line design and replacement, preparation of construction drawings and specifications, and coordination with multiple City departments.

Significant utility crossings will include fiber optics, sanitary sewer, and sensitive aging water lines. Permits were obtained from railroads for jacking and boring storm sewer and water lines under three railroads.

This project included several unusual water quality elements. These included four large water quality vaults called Carlsonators, which use an extensive travel time to maximize the settlement of smaller suspended solids.

Three additional water quality features were also designed: a bio-box, which is a small bio-retention system; a bio-basin, which also acts as a filtration system; and a modified storm inlet, which allows flows from smaller storm events to infiltrate into the soil below the inlet. These three water quality features are experimental and were jointly designed by the City and Ayres Associates.

Ayres Associates staff previously completed the design of Phases 1, 2 and 3 of the Washington Avenue drainage system project.

Client Name: City of Loveland
Address: 2101 O'Neil Avenue, Cheyenne, WY 82001

Contact Person: Christopher Carlson
Telephone: 970.962.2774

Project Dates: May 11, 2007 - October 1, 2011

Original Project Budget: \$3,400,000

Final Project Cost: \$3,085,000

Explanation of Variation from Original Budget to Final Project Cost: The above costs are total project costs. The design budget was \$210,000. The actual cost was less because a portion of the project was built during a later phase.

Canal Importation Ponds and Outfall (CIPO) Design and Construction Phase Services



Client Name: City of Fort Collins
Address: 700 Wood Street, Fort Collins, CO 80521
Contact Person: Matt Fater, PE
Telephone: 970.416.2146
Project Dates: July 31, 2006 - July 1, 2011
Original Project Budget: \$1,378,296
Final Project Cost: \$1,378,296
Explanation of Variation from Original Budget to Final Project Cost: This was a Negotiated Scope Contract of \$1.3 million for design. The construction budget was \$20 million; \$4 million/year for five years.

The largest stormwater project undertaken by the City of Fort Collins since the 1997 flood involved design and construction of \$20 million of stormwater improvements in heart of the City directly upstream of Colorado State University (CSU). The project included extensive public outreach, including mailings, open houses, and a project website to educate the public on the project's impact on neighborhood roadways and primary traffic corridors serving CSU. The project involved reducing spills from three drainage ditches that affected more than 175 homes during a potential 100-year storm event. The project included complex hydrologic and hydraulic modeling (using SWMM and unsteady HEC-RAS) to determine ditch spill quantities and mapping of the existing floodplain.

The project included design of five regional detention/water quality ponds and more than 4,000 feet of large diameter (up to 102 inches) storm sewer and associated laterals and inlets. Inlets include use of Snouts and upstream "sweeper sections" to improve water quality in downstream receiving waters. The project included design of a round, aesthetically enhanced version of the Bureau of Reclamation Type 10 hanging baffle structure. Ayres Associates provided construction services through July 2011. The Colorado Association of Stormwater and Floodplain Managers honored the project as the 2012 Grand Award for Engineering Excellence winner.

Conflict of Interest

None of our current clients, or those of our subconsultants, have interests that could compete or conflict with the City of Cheyenne's Lake Minnehaha Remediation Plan Phase 2 project.



Lake Minnehaha Remediation Plan Phase 2
City of Cheyenne

Other Information



AYRES
ASSOCIATES

Ayres Associates has no additional information to include in this proposal.



January 16, 2015

Mike Vinson
City of Cheyenne, City Engineers Department
2010 O'Neil Ave., Room 206
Cheyenne, WY 82001

Re: Price Proposal
RFP 7-15 – Lake Minnehaha Remediation Plan Phase 2

Dear Mike:

Enclosed please find our price proposal for the scope of services defined in Request for Proposal 7-15. We have provided a cost breakdown for work proposed by Ayres Associates and our subconsultants, Steil Surveying Services and Terracon, based on the Scope of Work to complete all necessary items.

Ayres Associates, Steil Surveying Services, and Terracon have reviewed the City's Professional Services Agreement and are in agreement with the obligations contained therein. Our subconsultant agreements with Steil and Terracon will include flow-down provisions that obligate our subconsultants to the City's prime Professional Services Agreement with Ayres Associates.

As requested, this price proposal will be valid for at least 120 days.

The signers of this proposal shown below declare that the companies interested in this contract as principals are named herein, specifically Ayres Associates, Steil Surveying Services, and Terracon. This proposal is offered in all respects as a fair proposal, in good faith without collusion or fraud. James Schall, PhD, PE, has authority on behalf of Ayres Associates to bind the company at the contract value shown in the price proposal.

Thank you for considering our proposal.

Sincerely,

Ayres Associates Inc

Christopher G. Pletcher, PE
Project Manager

James D. Schall, PhD, PE
Vice President

Ayres Associates - Professional Services Fee Proposal
City of Cheyenne - City Engineer's Office
Lake Minnehaha Remediation Plan Phase 2
Based on RFP-7-15



LABOR DETAIL AND SUMMARY OF DIRECT AND SUBCONSULTING COSTS BY TASK

Task	Description	Project Manager (Chris Pletcher)	Technical Advisor (Andrea Faucett)	Project Engineer (Sam Lowe/Darci Hendon)	Design Engineer (Pat/Florian/Samantha)	CADD Technician (Rhonda Bunner)	Total Labor Cost	Direct Cost	Subconsultant Hours	Subconsultant Cost	Total by Task
		\$ 160	\$ 175	\$ 130	\$ 95	\$ 115					
1	Preliminary Design Phase										
1.1	Geotechnical Investigation (Terracon)	1		4	8		\$ 1,440	\$ -	50	\$ 8,150	\$ 9,590
1.2	Topographic Survey (Steil)	1					\$ 160	\$ -	237	\$23,700	\$ 23,860
1.3	Lake Minnehaha Preliminary										
1.31	Trail Realignment			8		12	\$ 2,420	\$ -		\$ -	\$ 2,420
1.32	Parking Lot Removal			3		2	\$ 620	\$ -		\$ -	\$ 620
1.33	Gazebo	16			12	4	\$ 4,160	\$ -		\$ -	\$ 4,160
1.34	Bio-bed	4		20	20	6	\$ 5,830	\$ -		\$ -	\$ 5,830
1.35	Flatter Bank Slopes		2	4	8	2	\$ 1,860	\$ -		\$ -	\$ 1,860
1.36	Landscaping	8		8			\$ 2,320	\$ -		\$ -	\$ 2,320
1.4	Storm Sewer Improvements										
1.41	Duff Avenue System / Inlet Capacity	16	4	20	80	80	\$ 22,660	\$ -		\$ -	\$ 22,660
1.42	Three Water Quality Structures		4	16		8	\$ 3,700	\$ -		\$ -	\$ 3,700
1.43	Decommissioning of Bypass Sewers		1	2	4	2	\$ 1,045	\$ -		\$ -	\$ 1,045
1.44	SWMM Capacity Analysis		4	20	20		\$ 5,200				
1.5	Preliminary Design Report	8	4	40	50	4	\$ 12,390	\$ -		\$ 5,080	\$ 17,470
1.6	Public Meeting	8		12		4	\$ 3,300	\$ -		\$ -	\$ 3,300
	Subtotal	62	19	157	202	124	\$ 67,105	\$ -	287	\$36,930	\$ 98,835
2	Final Design Phase										
2.1	Final Design and Specs	16	8	40	80	80	\$ 25,960	\$ -		\$ -	\$ 25,960
2.2	Final Cost Estimate	2		8	40		\$ 5,160	\$ -		\$ -	\$ 5,160
2.3	Project Manual			32			\$ 4,160	\$ -		\$ -	\$ 4,160
2.4	Construction Sequencing and Traffic Routing Plans			8	24	16	\$ 5,160	\$ -		\$ -	\$ 5,160
2.5	Public Meeting	8		12		4	\$ 3,300	\$ -		\$ -	\$ 3,300
2.6	Final Bid Package	8		8		4	\$ 2,780	\$ -		\$ -	\$ 2,780
	Subtotal	34	8	108	144	104	\$ 46,520	\$ -	0	\$ -	\$ 46,520
3	Bidding & Construction Phase										
3.1	Bidding	16		16		8	\$ 5,560	\$ 231		\$ -	\$ 5,791
3.2	Construction Engineering	20		80	268	8	\$ 39,980	\$ 1,520	24	\$ 2,400	\$ 41,500
	Subtotal	36	0	96	268	16	\$ 45,540	\$ 1,750	24	\$ 2,400	\$ 47,290
	Total	132	27	361	614	244	\$159,165	\$ 1,750	311	\$39,330	\$192,645

REQUEST FOR PROPOSAL No.: RFP – 7 – 15

PROJECT NAME: LAKE MINNEHAHA REMEDIATION PLAN PHASE 2

ATTACHMENT "B-3"
USE OF SUBCONSULTANTS

Sub-consultant Name/City/State	% of Value of Work	Work Tasks to be Assigned
Steil	14	Survey
Terracon	3	Geotechnical
Total Value of Subcontracts		

PROFESSIONAL SERVICES AGREEMENT

Project Name: LAKE MINNEHAHA REMEDIATION PLAN PHASE 2 (RFP-7-15)

EXHIBIT "B"

COMPENSATION SCHEDULE

The Consultant shall prepare and submit proposed estimated compensation, bearing in mind that it should be a unit-based contract with a negotiated not-to-exceed amount. A Payment Schedule based upon the following shall be submitted with payment for a specific phase not exceeding the scheduled amount prior to completion of that phase:

Preliminary Design Phase	\$ <u>98,835</u>
Final Design Phase	\$ <u>46,520</u>
Engineering Design Subtotal	\$ <u>145,355</u>
Bidding and Construction Phase	\$ <u>47,290</u>
Total Contract Amount	\$ <u>192,645</u>

PROFESSIONAL SERVICES AGREEMENT

Project Name: **LAKE MINNEHAHA REMEDIATION PLAN PHASE 2 (RFP-7-15)**

EXHIBIT “C”

PROJECT SCHEDULE

The key milestones in the Lake Minnehaha Remediation Plan - Phase 2 project, including storm sewer construction, and water & sewer improvements are shown below. **The proposer should complete the following schedule as part of the proposal and include any other key dates needed.**

Date	Schedule Item
March 11, 2015	Notice-to-proceed issued by City
<u>March 17, 2015</u>	Hold scoping meeting with City Staff
<u>July 20, 2015</u>	Submit preliminary design and report
<u>April 16, 2015</u>	Meet with City to discuss review of design
<u>Aug. 4, 2015</u>	First Public Meeting
<u>July 20, 2015</u>	Submit 65% design documents
<u>Aug. 3, 2015</u>	Meet with City to discuss review of design
<u>Sept. 23, 2015</u>	Second Public Meeting
<u>Sept. 9, 2015</u>	Submit 95% design documents
<u>Sept. 22, 2015</u>	Meet with City to discuss review of design
<u>Oct. 6, 2015</u>	Final design submittal and review
<u>Nov. 18, 2015</u>	Pre-bid meeting
<u>Nov. 25, 2015</u>	Bid Opening

* For a detailed proposed schedule for the Lake Minnehaha Remediation Plan - Phase 2 project, please see Ayres Associates' Gantt chart on the following page.